ClawBench: Can AI Agents Complete Everyday Online Tasks?

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Research Questions

- Can current AI agents complete realistic everyday online tasks on live production websites?
- How do we safely evaluate agents on real websites while preventing unwanted side effects?
- What is the performance gap between existing sandbox benchmarks and real-world web tasks?

Core Idea

Existing web agent benchmarks test on simplified sandboxes with static HTML, missing the complexity of real websites (dynamic content, authentication, anti-bot defenses). ClawBench evaluates agents on 153 everyday tasks across 144 live production sites by intercepting only the final HTTP submission request that would commit transactions, preserving real-world complexity while preventing side effects. The key insight is using human reference trajectories and an agentic evaluator for comparative assessment rather than isolated task evaluation.

Key Formula

Agentic Evaluator Score

$$Score(t) = A(q^{(t)}, T_a^{(t)}, T_h^{(t)})$$

Score(t) ∈ {0,1} is binary task completion verdict, q^{(t)} is task instruction, T_a^{(t)} is agent trajectory, T_h^{(t)} is human reference trajectory, A is the agentic evaluator that compares agent against human performance

Success Rate

$$SR = \frac{1}{|T|} \sum_{t \in T} Score(t)$$

SR is overall success rate, |T| is number of evaluated tasks, summation aggregates binary scores across all tasks to compute percentage completion rate

Key Figure

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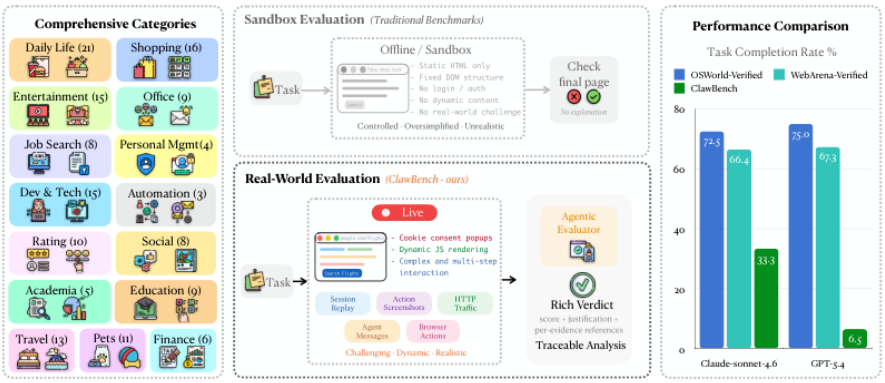

Figure 1: CLAWBENCH overview. **Left**: 153 tasks across 15 life categories. **Middle**: existing benchmarks evaluate agents in offline sandboxes with static HTML and fixed DOM structures; CLAWBENCH evaluates on live websites with real-world complexity and provides rich, traceable verdicts via an agentic evaluator. **Right:** Claude-Sonnet-4.6 and GPT-5.4 achieve 65-75% task completion on established benchmarks such as OSWorld and WebArena, but only 33.3% and 6.5%, respectively, on CLAWBENCH, highlighting the difficulty of real-world everyday web tasks.

Figure 1: Shows the dramatic performance gap where Claude Sonnet 4.6 achieves 65-75% on traditional benchmarks but only 33.3% on ClawBench, highlighting the difficulty of real-world tasks.

Limitations

- Limited to 153 tasks across specific categories, may not cover all everyday web interactions
- Relies on human-annotated interception signals which requires manual expertise and may miss edge cases
- Evaluation depends on agentic evaluator (Claude Code) which could introduce bias or inconsistencies
- Real websites change over time, making benchmark maintenance challenging and results potentially non-reproducible